

Churn Prediction & Player Behaviour Analysis on a Subscription Based Online Game

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Abstract

With tens of TBs of data being generated every day, data mining has become essential to gain an upper hand over other competitors. This alludes to the problem faced by this research, being how can a subscription-based game retain their players from churning and going over to numerous other games available to them. The solution will provide a system that predicts the player's chance to churns in the coming months and an in-depth analysis on the players' behaviour, highlighting those features which are important to the players. Hence, assisting in identifying and retaining those players which are about to churn.

In this project, two questions will be tackled, being 'when' and 'how' will a company direct business strategies towards a player who is starting to decline in an online subscription-based game. This will be accomplished by applying data mining techniques on an MMO dataset from the game, World of Warcraft (WoW). The 'when' part of the research will be addressed using different prediction models, while defining the suitable features and data look-back used by the model. This will provide a clearer view of when the company need to start their marketing strategies. The 'how' problem will be tackled by grouping players with similar interests together using clustering techniques and, by analysing patterns and relationships within the features of each group, targeting the users with more personalised marketing strategies.

1. INTRODUCTION

Churn prediction and customer behaviour analysis are frequently used in the industry and are vital for a company's road to success. Hence, it stands to reason that these research areas are well documented in both videogame-based and non-videogame-based businesses. Different approaches to predict customer behaviour imply that different algorithms are utilised depending on the data and aim of the research [1].

Massively multiplayer online (MMO) games are a type of videogame that is played online with a large number of players, often simultaneously. Being the most played MMO of the year [2-4], catering for around 120 million players this year alone, World of Warcraft (WoW) [5] was a good contender for this project.

With video games generating close to 50TB of data every day [7][8], dealing with big data proves to be beneficial. However, data telemetry alone is not enough, data analysis is what gives meaning to the raw data obtained and is what deems it advantageous. This amount

of data signifies that there is an overabundance of games for the players to choose from. Getting the player to initially buy the game is already a great feat, however, especially in subscription-based games, being able to retain the player is as important. The data mining techniques provided in this paper will provide a means to combat this issue by providing a way to know when the player is about to churn and outline means to analyse important features of the user's behaviour, to gain a better understanding of what the player plays the game for.

Some of these MMOs release their datasets publicly or provide APIs to get real-time data, allowing the public to create their own dataset. This project makes use of the latter option, using a dataset which was created using API calls, querying the servers every ten minutes.

Knowing when the player is about to churn and how can they be persuaded to be retained is tremendously important to a business. Prediction models such as SVMs, regression, deep learning etc. is no new feat, however, as can be seen further on, there is no right way to approach this problem, since the model to be created is heavily dependent on the data provided. These models can be used for prediction in short-term games, using little to no pre-dated data while also on others, such as WoW with subscriptions based on 1, 3 and 6 months, that can have more leeway.

Tackling the research question '*how can a player be persuaded to continue his subscription when they are predicted to churn?*' depends on a number of features - being aware of certain patterns in the data will create a deeper understanding of the player's behaviour. Furthermore, players can be clustered into groups of similar interests, being able to cater for these players with personalised marketing strategies.

2. AIMS AND OBJECTIVES

The main aim of this project is to create a model that can predict player churn in the following months and analyse player behaviour to gain a clear understanding of the user's actions in the game, hence being able to identify when and how can companies confront these users to try retaining them.

To achieve this aim, the following objectives have been set:

- Investigating different models capable of predicting player churn, finding the one which suits the dataset best.
- Finding the optimal amount of pre-dated data the model needs to precisely predict player churn.

- Analysing which features can be used for churn prediction and clustering, their relevance, and their relations with each other, highlighting those features that excel the most.
- Clustering the players into groups depending on behaviours and characteristics, underlining how they differ from each other and how different features stand out in each group of players.

3. LITERATURE REVIEW

The topics of churn prediction and customer behaviour analysis are frequently used in the industry and are vital on a company's road to success, hence why these research areas are well documented in both videogame-based and non-videogame-based businesses. Different approaches to reach a similar objective imply that different algorithms are utilised depending on the data and aim of the research [1].

The datasets used by similar research will first be discussed, followed by a discussion of how similar systems from literature tackled the objectives that were set out in Section 2.

3.1 Dataset

Choosing the dataset provides the foundation needed to define what objectives can be tackled for research. Apart from the raw data itself, proper data usage and manipulation of the dataset is what produced meaningful data, by using suitable data mining techniques. Finding the right dataset could be troublesome since most of the AAA games keep this data private for personal use to not give any edge over competitors.

The World of Warcraft Avatar History Dataset [9], the dataset used for this project, and used as a foundation by other researchers [10-12], contains data about the players in increments of 10 minutes for 3 years, dating from 2006 to 2009, with over 90,000 avatars and 150,000 observations. Containing a plethora of logs, it is imperative that this data is worked upon, with [10] and [12] tackling this problem by accumulating the 10-minute intervals into days, while [11] goes a step above and accumulates this to months. Different approaches to manage the size of the data were also considered, the dataset authors in [10], used the whole 3 years of data, while on the other hand, Siqueira et al. [11] used data from 2008 only since a popular update was released during the year. Nery et al. [12] also uses the whole 3 years but addresses this problem by testing the model for each separate year and also with the whole 3 years of data.

One can find other similar datasets that have been used in literature. For instance, the datasets used by Drachen et al. [13], Yang et al. [15] and Zhou et al. [24] are all free-to-play games, where all datasets contained over 100,000 users. Facing the short lifetime issue, all 3 games only consider those users that played the game for

a certain amount after installation, these being 7 days [13], 15 days [15] and 21 days [24]. It was also highlighted how all the users considered playing the same version of the game to eliminate any bias [15][24]. Tamassia et al. [1] compile their own dataset, where data containing over 10,000 players is taken from the game *Destiny*. Similar to [10-12], the data is collected as daily aggregates, which is accumulated to represent weeks.

3.2 Churn Prediction

Apart from in-app purchases, subscription-based products, such as 'World of Warcraft' depend on retaining the customer, in this case, the player, for a prolonged time, using marketing measures to do so. Thus, knowing when the player is likely to stop playing the game and henceforth discontinue his subscription is profitable to the company. In this section, different algorithms, namely SVMs, regression, Decision Trees, Random Forests, Naïve Bayes and Neural Networks used for churn prediction on both gaming and non-gaming based datasets will be discussed.

Support Vector Machine is a standard classification algorithm which is considered by most as a benchmark algorithm for classification problems, classifying a player as a churning or not in this particular case. The WoWAH dataset authors [10] use this technique to predict churning in 3 to 60 days, obtaining results with up to 85% accuracy. Rodan et al. [16] showed how SVMs are effectively used in businesses outside the gaming industry, achieving up to 98.7% accuracy by predicting 20 days into the future. Drachen et al. [13] compared their work on short-term free-to-play games using a decision tree to an SVM, apart from other algorithms, producing decent results with up to 79% accuracy. This showed potential in predicting with 1 session, 1 day and 1 week of pre-dated data to predict player churn in the next week.

Regression is one of the most famous types of algorithms used for data prediction, proposed by Siquiera et al [11] as an implementation in the gaming industry while others [1][13][14] compared their implementations to the said algorithm. Siquiera et al's [11] implementation of the algorithm uses the same dataset and similar objectives to those proposed by this work, being churn prediction, player clustering and player behaviour analytics, hence serving as a foundation for the work proposed by this paper, producing a model which returns a percentage change of churning or retaining. From compared implementations, Hadiji et al. [14] and Tamassia et al's [1] regression models compared quite decent, placing second for the former and producing the highest F1 score in [1]. On the other hand, results were average in Drachen et al. 's [13] implementation.

Drachen et al. [13] make use of Pruned Decision Trees in free-to-play mobile games because of their simplistic nature. This facilitates rapidly building a model as the player installs the game, to promptly predict the player's possibility of churning and hence, act accordingly with

marketing efforts. The main reason behind this is because of the fact that these types of games tend to have a short lifetime of a few days [13-15]. Drachen et al. [13] illustrate this by only allowing the decision tree to contain 3 to 4 decision rules which seemed to be enough when taking into consideration a few play sessions. This resulted in scores of up to 72% accuracy.

Decision trees are also the main research area of Hadiji et al. [14], which had a different approach to the prediction problem. Rather than having set a cut-off date as others have done [10][11][13][16], 1 to 2 days are set to compensate for the slow decline of a player. When predicting for 7 days, the model performed with an accuracy of 92%. On the other hand, when compared to other models by [1], decision trees acted poorly in every aspect.

Random Forests of Trees model's performance was notable when compared to by other models in research papers [1][13], concluding with the best or near best results.

Mishra and Reddy [17] make use of deep learning for churn prediction, in the form of a Convolutional Neural Network. Three layers are used, being the input, one hidden layer and the output, using the sigmoid function as an activation function. This resulted in a model with an 86% accuracy. Other literature [16][14] compare their work to Neural Networks, with results being moderately well, but still overshadowed by other techniques.

Tamassia et al. [1] conclude that churn prediction can be classified as a Hidden Markov Property problem, by recognising certain patterns as classifiers of a churning or not. This had poor accuracy results but performed well in the precision score.

Naïve Bayes is another algorithm which can be used and is quite simple in nature, however, research on churn prediction using this specific technique in this scenario is quite limited. [1][16][14] all used this algorithm to compare their proposed work, resulting in the Naïve Bayes model acting poorly in every instance.

It can be noticed that most of the algorithms excel in their own aspects, which is why understanding their strengths and weaknesses is essential when choosing the model. This also depends on the data being used by the model itself. Hence, some more importance will be given to those which use the same or similar datasets.

From literature, regression models [1][13][11][14] seemed to be the most commonly used model for prediction, obtaining similar promising results to SVMs [10][16][13], however, Random Forests models seemed to achieve the best result in churn prediction, obtaining the highest or nearly the highest model scores [1][16]. Then again, some experiments with different models will be beneficial which may lead to finding some unexpected results.

3.3 Data Look-Back

Different games have different lifetimes and hence the amount of data used to predict player churn varies with each game, henceforth having the optimal amount of data look-back is crucial in obtaining the best results. Tarng, Chen, and Huang, who compiled the original dataset [9], experimented for the best amount from 2 to 20 days of preceding data, concluding that 10 days seemed to produce the best results, validating this with 10-fold cross-validation [10]. This was done to predict 1 month into the future, hence having a ratio of 1:3 days for look-back to prediction days. Drachen et al. in [13] used a similar ratio which used 1-3 days of preceding data to predict 4-10 days into the future, resulting in a 75% accuracy model.

Hadiji et al. in [14] highlighted how more than 95% of the player's absence time lie in the last 7 days before churning, which also signifies the full cycle of a week. Depending on the week cycle, the model will predict the player's behaviour in the next, mainly being the percentage that a player is retained or churned, hence having a 1:1 ratio with a week of preceding data and a week of prediction. W. Yang et al. [15] consider the game's life span as 15 days, dividing it into 5 sections of 3 days each. Here, the model considers the preceding 3 days of data and predicts the outcome of the next 3-day section, hence having a similar 1:1 ratio. W. Yang et al. [15] also underline how the last day before the prediction corresponds best with the results.

3.4 Feature Importance

Using the right features is crucial in achieving high model scores, considering those features that correlate well with churning and also not over scaling the dataset with high dimensionality which might in some cases stray further rather than put the model on track. Three main feature classes were identified, these being those that are session-based, playtime-based and user-based.

Hadiji et al. [14] and Drachen et al. [13] both make use of session-based features, which as the name implies, revolve around the sessions played by the user. Hadiji et al. in [14] use features such as *Number of Days*, *Average Playtime per Session*, *Average Time Between Sessions*, *Number of Sessions*, and *Current Absence Time*, with the latter 3 being the most frequented features by the decision tree model, which on the whole contained other features which aren't session-based. It should be noted that these features were evaluated with 10-fold cross-validation to ensure that the right features were chosen. Drachen et al. [13] used similar features, which are *Current Absence Time* and *Average Time Between Sessions*, using intersession features alone rather than the sessions themselves. These resulted in being negative factors in the 1-3-day prediction, but on the other hand, exceptionally positive in the 7-day prediction, which shows that different prediction timeframes could require different features to maximise efficiency.

Conversely, playtime-based features used in [13], *Average Moves*, *Average Stars*, *Maximum Level* and *Average Duration* proved to be positive in the 1-3 prediction period and negative in the 7-day prediction period. Hadiji et al. [14] also used playtime-based features, using another model to get a player's retention value, which did not feature much in the grand scale of things. Tamassia et al. [1] made particular use of *Mean Lifespan*, *KD Ratio* and *Activities Completed* in their Hidden Markov Property Model, giving them 1 of 3 values, being below, equal or above the rolling average, which is a different approach to the others [16][13][14][12], which hence can be considered as normalisation of data.

User-based features are also used by multiple which reveal details about the users themselves rather than their play style. Hadiji et al. [14] consider the spending power of the user, being if the user is a premium user and other in-game purchases details. This is also a measure of how invested the player is in the game, since they are using their earnings in your game. Drachen et al. [13] go beyond that and uses more personal data, such as *device type*, *geo-location*, etc. Nery et al. [12] use the same dataset used in this paper and made some interesting findings within the dataset relating to the *races*, *classes*, and *behaviours* in the game. Nery et al. [12] also use 'RapidMiner' [19], which is a tool for data analysis and data mining, which further strengthened their decision, along with aiding in some observations. Although not in a video-game environment, Rodan et al. [16] make use of user-based features such as the *number of calls*, *mobile data used*, etc.

It can be noted that the majority mentioned contained some features from each category, which shows the importance of each category. Even if the features are not directly related to the prediction, they can still aid in understanding the reason behind churning. However, playtime-based, and more noticeably session-based features seemed to be more beneficial.

3.5 Clustering Player Behaviour

Player behaviour clustering will be used to support two of the other objectives mentioned in Section 2, player churn prediction and feature importance. Clustering will be investigated as a merged approach with the prediction model, creating a model for each group, which may require different parameters for the different clusters, and may result in a more accurate system. Similar to the work done by Nery et al. [12], the grouping of players with similar interests will give a deeper understanding of the players, finding important features within each group, avoiding generalising with the whole player base.

Clustering algorithms vary when changing the slightest parameter, moreover, choosing one algorithm instead of another tremendously varies the results [11]. Hence, choosing the right one is crucial for finding the

desired result from the given dataset. There are three main types of clustering techniques, these briefly are [20]:

- *centroid models*, which revolve around the closeness of data points to the centroids allocated
- *distribution models*, which clusters data falling under the same distribution (e.g., Normal, Gaussian)
- *density models*, which revolve around clustering density regions within the dataset

Choosing the right algorithm depends on the dataset to be used and the desired results, since for example when using centroid algorithms, one needs to have prior knowledge about the dataset to allocate the number of clusters. Different clustering approaches are also possible, those where data is exclusive to one cluster, known as Hard Clustering and on the other hand, a fuzzy-logic approach where data can be partially assigned to multiple clusters, known as Soft Clustering [21].

K-Means Clustering is one of the most commonly used centroid algorithms to date, being an easy algorithm to implement because of all the tools and their documentation being easily accessible and producing results for general purposes. Siqueira et al. in [11] briefly highlight how K-means, Principal Components Analysis (PCA), Non-negative Matrix Factorization (NMF) and Archetype Analysis (AA) tackled the problem of dimensionality reduction, deciding to make use of k-means since it fit the dataset better. The players in the dataset were clustered based on their in-game levels into 4 clusters on Playtime vs Playing Density in equally distributed player-level increments. It should be noted that [11] is using the same dataset to be used in this paper. They found that 50% of the WoW players play longer than 48 hours per month, which fits the middle two clusters. Moreover, the upper two clusters, making up to 25% of players, play longer than 84 hours per month.

Bauchhage, Drachen and Sifa [22] experimented with different clustering algorithms, concluding that Archetype Analysis (AA) clustered on high or low distinctive scores while k-means provided more broadly defined clusters. Some key concerns regarding clustering were also underlined, these were:

- having the right validation tools
- informative data which can be interpreted and visualised
- time and progress dependencies within the dataset
- high dimensionality and size of data
- different data types
- choosing the right features

They also suggested that initial centroids should at least be selected among the available data points rather than the default random initialisation, because of the fact that letting the algorithm choose which data points to start as centroids will result in different clusters every execution. It should also be noted that K-Means only finds the local optimal solution and is near-impossible to

find the global optimal solution in reasonable time. Bauckhage, Drachen and Sifa [22] also recommended that for general models, centroid models work better, on the other hand, models such as AA should be utilised for clustering the extremes in the dataset. AA should also be used in high-dimensionality data sets, where data might be too sparse for centroid clustering to group data points together. They also stated that Density-based algorithms should be used in instances where the data is too tightly grouped for centroid algorithms to find clusters.

Yang et al. [23] used K-Means clustering to group players based on their buying power into 3 groups ($k = 3$), choosing from 100 instances, up to 5000 players, which were tested using their silhouette scores and A/B testing with a benchmark algorithm.

Zhou, Hu, and Liu [24] use clustering based on time series, being represented in days. The problem of sparsity was encountered, since the players did not play every single day, resulting in having a number of zero values. It was decided to set k to 6, which turned out quite useful since they found how different classes tend to die out, some being immediate while others at a slower pace. This can help to cater for different marketing strategies depending on which cluster a player falls under. The data was then visualised using two plots, which are Number of Rounds Vs Days and Coins Vs Days, which overcome one of the concerns presented by [22] earlier.

As one could conclude, K-Means Clustering seems to be an optimal solution for a general centroid base algorithm and is what seems promising for this project, however, the number of centroids varied throughout the literature. This denotes that one needs to have at least a vague idea of the desired results and act similarly to [10], experimenting with a different number of centroids around the ballpark of the desired value, and cross validating each finding.

4. PROPOSED METHODOLOGY

The proposed solution will be divided into four sections, being the four objectives discussed in Section 2. Each of the subsection will discuss and evaluate results produced by literature and conclude on which systems look promising to be used in this project.

4.1 Churn Prediction

The first objective covers the actual model used for prediction. As seen from Section 3.2, there are numerous approaches to address this problem, hence different models will be tested and compared to each other to see which ones suit the WoWAH dataset [9] best. The models that will be evaluated are Regression and SVMs, both yielding relatively high results while also using the results of Tarnig et al. [10] and Siqueira et al. [11] both as a foundation and as a comparison since they use the same dataset for churn prediction. The Random Forest algorithm will also be evaluated, based on the promising

results found in literature [1][13]. To compare results based on the extremes in complexity of the system, both Decision Trees and Neural Networks will be evaluated. The 1 and 3 months of prediction were selected based on WoW's subscription options, being 1, 3, and 6 months, opting for the former two since 6 months might be too far into the future in practice.

This will result in a model/s (since different models might be suitable for the different prediction windows) that given certain parameters about the player will yield the percentage chance that the player churns.

4.2 Data Look-Back

This section underlines the amount of pre-dated data that needs to be collected by the player to accurately predict his departure. From literature, the most common look-back to prediction ratios seemed to be 1:1 [14][15] and 1:3 [10][13], which are two ratios that will surely be evaluated. However, most of these research papers make use of games with a short lifespan, hence why different ratios such as 2:1 are also going to be tested based on the long-term lifespan and available data of WoW.

4.3 Feature Importance

Knowing which features represent the player best, how they fluctuate with different players and how they correlate with each other will give the developers a deeper understanding of the player base. These features are also useful in Sections 4.1 and 4.4, utilising useful and informative features for churn prediction and clustering. As discussed in Section 3.4, *Session-based*, *Playtime-based* and *User-based* features were utilised together by the majority of the literature, indicating that some features from all categories should be used. However, not all categories will have the same weight. User-based features, being more informative rather than having a high correlation with both models, will have less weight than the rest. Conversely, playtime features [13][14] such as *playtime duration* and *player level*, and more over session-based features [1][13][14] such as *number of sessions* and *absence time between sessions* are features that showed high correlation with both prediction and good contenders in clustering, which seemed promising to use in this project.

4.4 Clustering Player Behaviour

Clustering of players behaviour involves grouping players with similar interests and characteristics together, avoiding generalising with the whole player base. Cluster of players will help identify and analyse important features in each sub-group; revealing different player patterns, whose behaviour may vary greatly between groups [12]. Different player behaviour also alludes to different churn patterns, which may require different models for each group to produce a more accurate result. Based on the findings of [22][23][24][9], the proposed algorithm to be used will be K-Means Clustering,

evaluating the best features and model parameters to be used.

5. EVALUATION

This system will be evaluated using the WoWAH dataset [9] (described in Section 3.1) by means of two widely used types of scoring metrics, these being Accuracy and F1 Score. Accuracy signifies the number of correct predictions for every prediction made by the model itself. It is one of the most utilised evaluation metrics found in research, being used by [1][10][16][13]. F1 score merges, using harmonic mean, Recall and Precision metrics which focus more on the accuracy of the positive cases. Similarly, to Accuracy, this metric was also used frequently in research papers, used by [1][13][14][9]. To evaluate and validate the features used in both the prediction and clustering models, k-fold cross-validation will be used, which evaluates the features on sub-sets of the dataset. The most common value for k is 10 [10][13][14], but 5 [16] and 3 [12] are also used.

6. CONCLUSION

The result of this project provides a model that is able to accurately predict if a player will churn in the next 1 and 3 months, given certain features as inputs. A deeper understanding of the player base will be achieved by analysing patterns and relationships within the features while also clustering the players into groups, evaluating the differences in interests and patterns in each group.

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8. GANTT Chart

Work Plan - GANTT Chart

